
EXPERIENCE

Senior Research Engineer

Dec. 2023 – Present

*Fujitsu Research of America, Inc.**Santa Clara, CA*

- Designed and trained a sequence generation model for predictive process monitoring. *[Python, PyTorch]*
- Engineered an ML-driven web application for business-process data analysis and visualization, **reducing manual effort (FTE) by 15%** through automated insight generation and streamlined reporting workflows. *[Python, LangChain, Scikit, Streamlit]*
- Automated process guidance by engineering a privacy-focused **multi-agent hierarchical workflow** using LangGraph and FastAPI. *[Python, LangSmith, vLLM, Docker]*
 - * Integrated RAG and self-reflecting coding agent for hybrid data analysis, achieving a **top-4 overall ranking** on the TableBench leaderboard. *[Chroma]*
 - * Implemented long/short-term memory to manage distinct conversation threads, delivering a full-stack solution deployed with Docker and Streamlit. *[PostgreSQL]*

Machine Learning Intern

Jun. 2021 – Dec. 2023

*Emplifi**Columbus, OH*

- Designed and developed a **knowledge-grounded response generation framework** to automatically answer user queries in a customer care setting (Real-world data) and deployed it as an API service. *[AWS, Docker, Python, PyTorch, HuggingFace, FastAPI]*
 - * Employed neural information retrieval (IR) techniques to collect knowledge from historical unstructured data and blended that information into generated responses through a fine-tuned GPT-2 model.
 - * The model generates **12% more informative and 15.3% more semantically similar responses** than that of vanilla GPT-2 (contemporary state-of-the-art).
- Developed a **paraphrasing model and corresponding web service** to introduce diversity in the business response template. *[AWS, Docker, Python, PyTorch]*

Graduate Research Associate

Aug. 2021 – Dec. 2023

*The Ohio State University**Columbus, OH*

- Designed a **hybrid ranking system** (sparse + dense) for product-QA RAG tasks, improving MRR@5 by 10.95% over lexical baselines. (Code, Dataset: [hetPQA](#)). *[Python, PyTorch, BERT, T5, FAISS]*
 - * Achieved 30% lower latency and 38% fewer FLOPs than SOTA cross-encoders with comparable accuracy.
- Designed and developed an **explainable deep-learning model - TransICD** to predict ICD codes from a patient discharge summary (Code, Dataset: MIMIC-III). *[Python, PyTorch]*
 - * The model achieves stable **micro-AUC and micro-F1 scores of 0.92 and 0.64** respectively with 1.1% and 2.5% increases over the corresponding metrics of LEAM (contemporary state-of-the-art).
- Designed and developed TransDTBA, a transformer-based model to predict drug-protein binding affinity (Code, Dataset: KIBA). *[Python, PyTorch]*
- Implemented joint slot-filling and intent detection, developed end-to-end NLG system for E2E NLG Shared Task (Code, Dataset: ATIS, E2E). *[Python, PyTorch, RNN]*

Graduate Teaching Associate

Aug. 2018 – May 2021

*The Ohio State University**Columbus, OH*

- Instructor of “Introduction to Computer Programming in C++” (Spring’20, Summer’20, Autumn’20 and Spring’20)

Engineer, Research and Development

Mar. 2016 – Jul. 2018

*KONA Software Lab Limited (in collaboration with KONA I, Seoul, Korea)**Dhaka, Bangladesh*

- Developed the front-end of a leading **Android payment application** in Bangladesh [NexusPay](#) (**5M+** Downloads)
 - * Utilized REST APIs for backend communication and implemented various transaction interfaces including NFC, QR-code, and online payment. *[Android, Java, SQLite]*
- Developed a Nexgo POS application to facilitate transactions from an Android payment application. *[C]*

EDUCATION

The Ohio State University

Aug. 2018 – Dec. 2023

Ph.D. in Computer Science and Engineering

Columbus, OH

- Advisor: [Dr. Rajiv Ramnath](#)
- Research Topics: Natural Language Processing, Text Generation, Information Retrieval
- Dissertation: [Improving Response Automation for Customer Communication through Document Categorization and Information Retrieval](#)
- Courses Taken: Machine Learning, Data Mining, Parallel Computing, Neural Networks, Speech and Language Processing, Social Media and Text Analysis
- GPA: 3.89 / 4.00

The Ohio State University

Aug. 2018 – May. 2023

Master of Science in Computer Science and Engineering

Columbus, OH

- GPA: 3.89 / 4.00

Bangladesh University of Engineering and Technology

Feb. 2011 – Mar. 2016

Bachelor of Science in Computer Science and Engineering

Dhaka, Bangladesh

- GPA: 3.73 / 4.00

PROFESSIONAL SERVICES

- Reviewer: EMNLP 2023 - Workshop BLP, ASONAM 2023
- Led a team of graduate students to employ a **deep reinforcement learning model for a question-answering task** in customer care service.

TECHNICAL SKILLS

Languages: Python, Java, C, C++

Frameworks & Others: PyTorch, LangChain, LangGraph, Streamlit, FastAPI, Scikit, spaCy, MCP, vLLM

Platforms: Android, AWS, Docker, CUDA, Databricks

Tools: Git, Jira, VS Code, Ollama, LangSmith, Cursor, Claude Code

LLMs: BERT, GPT, Gemini, Llama, Qwen, OpenAI APIs

Database: Oracle, MySQL, PostgreSQL, Chroma, Pinecone

PUBLICATIONS

- [Efficient and Interpretable Information Retrieval for Product Question Answering with Heterogeneous Data](#), Biplob Biswas, Rajiv Ramnath. In Proceedings of the Seventh Workshop on e-Commerce and NLP @ LREC-COLING, May 2024.
- [Retrieval Based Response Letter Generation For a Customer Care Setting](#), Biplob Biswas, Renhao Cui, Rajiv Ramnath. In Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL): Human Language Technologies: Industry Track, July 2022.
- [TransICD: Transformer Based Code-wise Attention Model for Explainable ICD Coding](#), Biplob Biswas, Hoang Pham, Ping Zhang. In International Conference on Artificial Intelligence in Medicine (AIME) Jun 2021.
- [Towards Simulating Non-lane Based Heterogeneous Road Traffic of Less Developed Countries](#), Q. M. Alam, B. Sarker, Biplob Biswas, K. H. Zubaer, T. Toha, N. Reza and ABM A. Al Islam. In ICT4S, May 2018.

HONORS AND AWARDS

- Dean's Merit List in 2nd and 4th year of Bachelors 2013 - 2016
- Education Board Scholarship - for notable performance in SSC and HSC examination 2010

ADDITIONAL INFORMATION

- H-1B visa approved in 2024